

Q1

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt/assignment_id=426&view=1`. The lab header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", the subject "JAVA PROGRAMMING", and the user's name "YAMANURI PRANEETH" with ID "1923722698A".

The main workspace is divided into three sections:

- Salary:** A text input field containing "39500.0".
- Your Code:** A code editor showing Java code for an employee management system. The code includes a `Main` class with a `main` method that uses a `Scanner` to take input and creates an array of `Employee` objects. It includes logic for different employee types: `TeachingFaculty`, `TechnicalStaff`, and `VisitingFaculty`.
- Input:** A text input field containing "3".
- Output:** A text area showing the output of the program, including "Employee Details" and "Employee ID : 101".
- Visible Test Cases:** A section showing three test cases, all of which are passed (indicated by green checkmarks).

The bottom of the screen shows a Windows taskbar with various application icons and system information, including the date and time (10:33 AM, 05-08-2026).

SIMATS Online Lab

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Ask Gemini

SIMATS ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

YAMANURI PRANEETH 1923722698A

```
194 private int employeeId;
195 private String employeeName;
196 private String department;
197 private double basicSalary;
198
199 public Employee() {
200 }
201
202 public Employee(int employeeId, String employeeName, String department, double basicSalary) {
203     this.employeeId = employeeId;
204     this.employeeName = employeeName;
205     this.department = department;
206     this.basicSalary = basicSalary;
207 }
208
209 public int getEmployeeId() {
210     return employeeId;
211 }
212
213 public void setEmployeeId(int employeeId) {
214     this.employeeId = employeeId;
215 }
216
217 public String getEmployeeName() {
218     return employeeName;
219 }
220
221 public void setEmployeeName(String employeeName) {
222     this.employeeName = employeeName;
223 }
224
225 public String getDepartment() {
226     return department;
227 }
228
229 public void setDepartment(String department) {
230     this.department = department;
231 }
232
```

198 return getBasicSalary() + (teachingHours * incentivePerHour);
199 }
200 }
201
202 class TechnicalStaff extends Employee {
203 private int overtimeHours;
204 private double overtimeRate;
205
206 public TechnicalStaff(int employeeId, String employeeName, String department, double basicSalary, int overtimeHours, double overtimeRate) {
207 super(employeeId, employeeName, department, basicSalary);
208 this.overtimeHours = overtimeHours;
209 this.overtimeRate = overtimeRate;
210 }
211
212 @Override
213 public double calculateSalary() {
214 return getBasicSalary() + (overtimeHours * overtimeRate);
215 }
216 }
217
218 class VisitingFaculty extends Employee {
219 private int lecturesHandled;
220 private double paymentPerLecture;
221
222 public VisitingFaculty(int employeeId, String employeeName, String department, double basicSalary, int lecturesHandled, double paymentPerLecture) {
223 super(employeeId, employeeName, department, basicSalary);
224 this.lecturesHandled = lecturesHandled;
225 this.paymentPerLecture = paymentPerLecture;
226 }
227
228 @Override
229 public double calculateSalary() {
230 return lecturesHandled * paymentPerLecture;
231 }
232 }

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Q2

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SIMATS Online Programming Lab

JAVA PROGRAMMING

YAMANURI PRANEETH 1923722698A

Your Code

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = readNextInt(sc);
8         Vehicle[] vehicles = new Vehicle[n];
9
10        for (int i = 0; i < n; i++) {
11            String type = readNextNonEmpty(sc);
12            String vehicleNumber = readNextNonEmpty(sc);
13            String ownerName = readNextNonEmpty(sc);
14            double insuredAmount = readNextDouble(sc);
15
16            if (type.equalsIgnoreCase("Car")) {
17                vehicles[i] = new Car(vehicleNumber, ownerName, insuredAmount);
18            } else if (type.equalsIgnoreCase("Bike")) {
19                vehicles[i] = new Bike(vehicleNumber, ownerName, insuredAmount);
20            } else {
21                vehicles[i] = new Truck(vehicleNumber, ownerName, insuredAmount);
22            }
23        }
24
25        String searchVehicle = readSearchVehicle(sc);
26
27        Vehicle highest = null;
28        double highestClaim = Double.NEGATIVE_INFINITY;
29
30        for (int i = 0; i < n; i++) {
31            Vehicle v = vehicles[i];
32            double claim = v.calculateClaimAmount();
33
34            System.out.println("Vehicle : " + v.getVehicleNumber());
35            System.out.println("Claim Amount : " + claim);
36            if (i != n - 1) System.out.println();
37        }
38    }
39}
```

Input

Car
TN08B1234
Ramesh

Output

Vehicle : TN08B1234
Claim Amount : 640000.0
Vehicle : TN08K15678
Claim Amount : 60000.0

Visible Test Cases

Test Case 1 ✓
Test Case 2 ✓
Test Case 3 ✓

25.00 / 25 marks

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JAVA PROGRAMMING

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```
11 for (Vehicle v : vehicles) {
12     if (v.getVehicleNumber().equalsIgnoreCase(searchVehicle)) {
13         System.out.println("Vehicle Found");
14         System.out.println("Owner : " + v.getOwnerName());
15         System.out.println("Claim Amount : " + v.calculateClaimAmount());
16         found = true;
17         break;
18     }
19 }
20
21 if (!found) {
22     System.out.println("Vehicle Not Found");
23 }
24
25 sc.close();
26 }
27
28 private static String readSearchVehicle(Scanner sc) {
29     while (sc.hasNext()) {
30         String token = sc.next().trim();
31         if (token.isEmpty()) continue;
32         if (token.equalsIgnoreCase("Search")) {
33             while (sc.hasNext()) {
34                 String t = sc.next().trim();
35                 if (t.isEmpty()) continue;
36
37                 if (t.equalsIgnoreCase("Vehicle")) { t.equalsIgnoreCase("Search"); continue; }
38                 if (t.toLowerCase().startsWith("vehicle:")) {
39                     String[] parts = t.split(":", 2);
40                     if (parts.length == 2) return parts[1].trim();
41                 }
42                 if (!t.equalsIgnoreCase("Vehicle") && !t.equalsIgnoreCase("Search")) return t;
43             }
44         }
45     }
46     return null;
47 }
```

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```
109 private static String readNextNonEmpty(Scanner sc) {
110     while (sc.hasNext()) {
111         String s = sc.next();
112         if (s == null) continue;
113         s = s.trim();
114         if (!s.isEmpty()) return s;
115     }
116     return "";
117 }
118
119 private static int readNextInt(Scanner sc) {
120     while (sc.hasNext()) {
121         String s = sc.next().trim();
122         if (!s.isEmpty()) return Integer.parseInt(s);
123     }
124     return 0;
125 }
126
127 private static double readNextDouble(Scanner sc) {
128     while (sc.hasNext()) {
129         String s = sc.next().trim();
130         if (!s.isEmpty()) return Double.parseDouble(s);
131     }
132     return 0.0;
133 }
134
135 abstract class Vehicle {
136     private String vehicleNumber;
137     private String ownerName;
138     private double insuredAmount;
139
140     public Vehicle(String vehicleNumber, String ownerName, double insuredAmount) {
141         this.vehicleNumber = vehicleNumber;
142         this.ownerName = ownerName;
143         this.insuredAmount = insuredAmount;
144     }
145 }
```

Windows Search

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```
144 public double getInsuredAmount() {
145     return insuredAmount;
146 }
147
148 public abstract double calculateClaimAmount();
149 }
150
151 class Car extends Vehicle {
152     public Car(String vehicleNumber, String ownerName, double insuredAmount) {
153         super(vehicleNumber, ownerName, insuredAmount);
154     }
155
156     @Override
157     public double calculateClaimAmount() {
158         return getInsuredAmount() * 0.80;
159     }
160 }
161
162 class Bike extends Vehicle {
163     public Bike(String vehicleNumber, String ownerName, double insuredAmount) {
164         super(vehicleNumber, ownerName, insuredAmount);
165     }
166
167     @Override
168     public double calculateClaimAmount() {
169         return getInsuredAmount() * 0.60;
170     }
171 }
172
173 class Truck extends Vehicle {
174     public Truck(String vehicleNumber, String ownerName, double insuredAmount) {
175         super(vehicleNumber, ownerName, insuredAmount);
176     }
177
178     @Override
179     public double calculateClaimAmount() {
180         return getInsuredAmount() * 0.90;
181     }
182 }
```

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